Motivation for Narrow Floating-Point Numbers

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Guide to Floating Point Formats

fp32: Single-precision IEEE Floating Point Format

Range: ~$1e^{-38}$ to ~$3e^{38}$

Exponent: 8 bits | Mantissa (Significand): 23 bits

S	E	E	E	E	E	E	E	E	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M

fp16: Half-precision IEEE Floating Point Format

Range: ~$5.96e^{-8}$ to 65504

Exponent: 5 bits | Mantissa (Significand): 10 bits

S	E	E	E	E	E	M	M	M	M	M	M	M	M	M	M

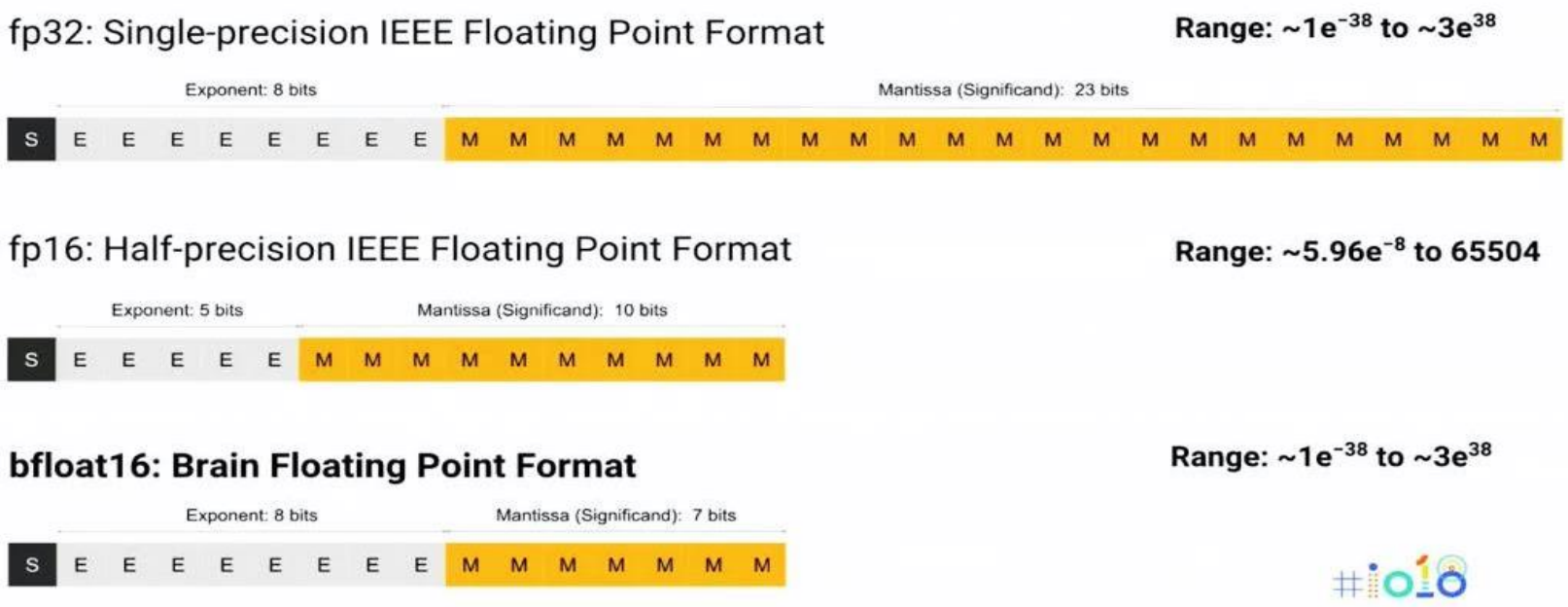
bfloat16: Brain Floating Point Format

Range: ~$1e^{-38}$ to ~$3e^{38}$

Exponent: 8 bits | Mantissa (Significand): 7 bits

S	E	E	E	E	E	E	E	E	M	M	M	M	M	M	M

FP16 vs Bfloat16



FP16 has only 5 exponent bits, hence, it has very low range

BFloat16 has 8 exponent bits, same as FP32. Hence, it can be used as a replacement of FP32 (although it has low precision)

Why is BFloat16 Better than Float16 for Model Training

TheAiEdge.io

We allocate the bits differently to built different Float numbers

Float 32

8 bits | 23 bits

1 0 1 0 1 0 0 0 1 0 1 1 0 0 0 0 1 0 0 0 1 1 0 1 0 0 0 0 0 1 1 0

Sign ← Exponent → ← Mantissa == decimal →

$$(-1)^{\text{sign}} 2^{(\text{exponent}-127)} (1 + \text{mantissa})$$

Float 16

5 bits | 10 bits

1 1 0 0 0 1 0 1 0 0 0 0 0 1 1 0

Sign ← Exponent → ← Mantissa →

$$(-1)^{\text{sign}} 2^{(\text{exponent}-15)} (1 + \text{mantissa})$$

Brain Float 16

8 bits | 7 bits

1 1 0 0 0 1 0 1 0 0 0 0 0 1 1 0

Sign ← Exponent → ← Mantissa →

$$(-1)^{\text{sign}} 2^{(\text{exponent}-127)} (1 + \text{mantissa})$$

Converting from Float32 to Float16 can overflow

Float 32

1 0 1 0 1 0 0 0 1 0 1 1 0 0 0 0 1 0 0 0 1 1 0 1 0 0 0 0 0 1 1 0

↓

Can float overflow

Float 16

1 ~~0 1 0~~ 1 0 0 0 1 1 1 0 0 0 0 1 0 0 ~~0 1 1 0 1 0 0 0 0 0 1 1 0~~

Converting from Float32 to BFloat16 is trivial

Float 32

1 0 1 0 1 0 0 0 1 0 1 1 0 0 0 0 1 0 0 0 1 1 0 1 0 0 0 0 0 1 1 0

↓

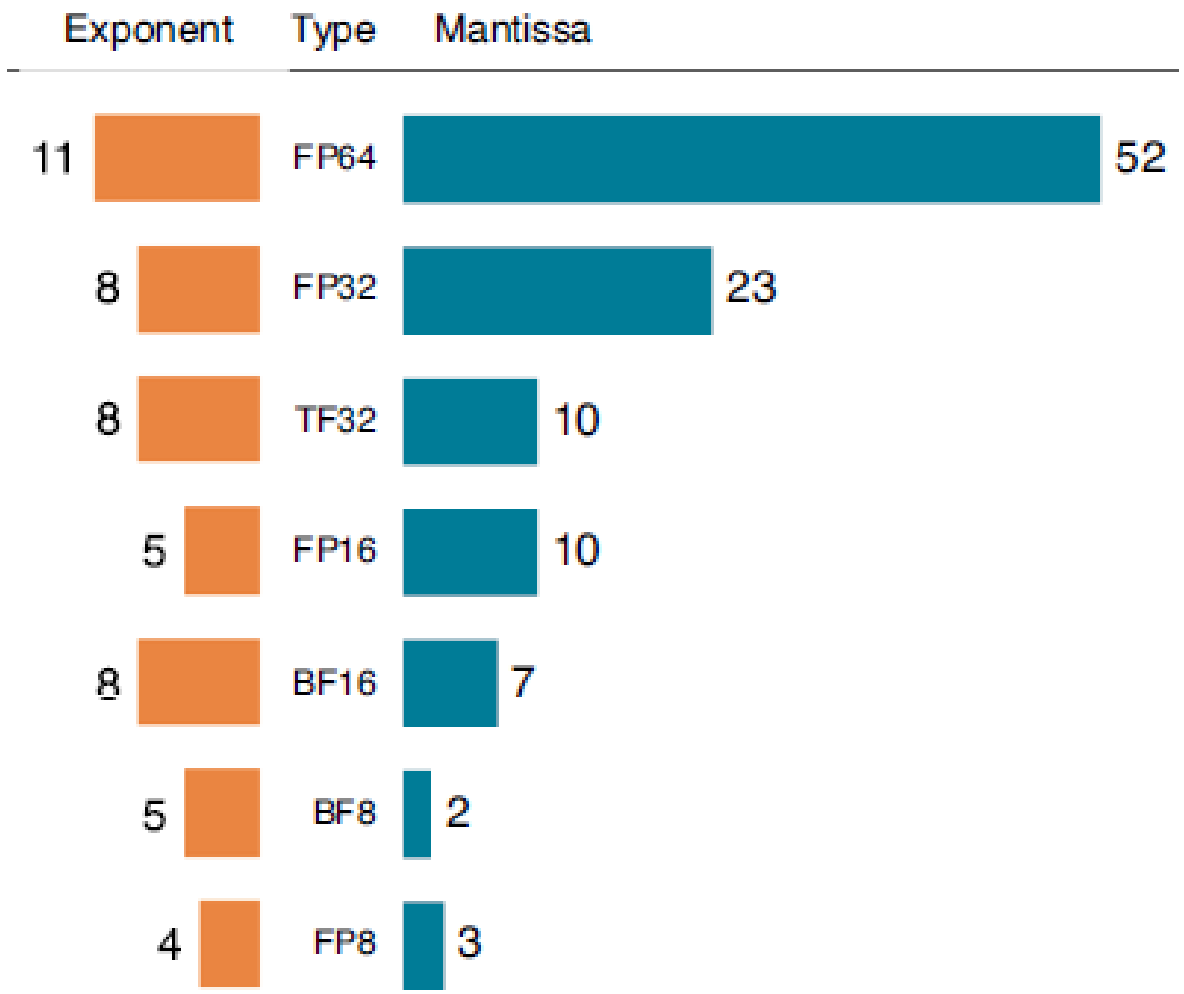
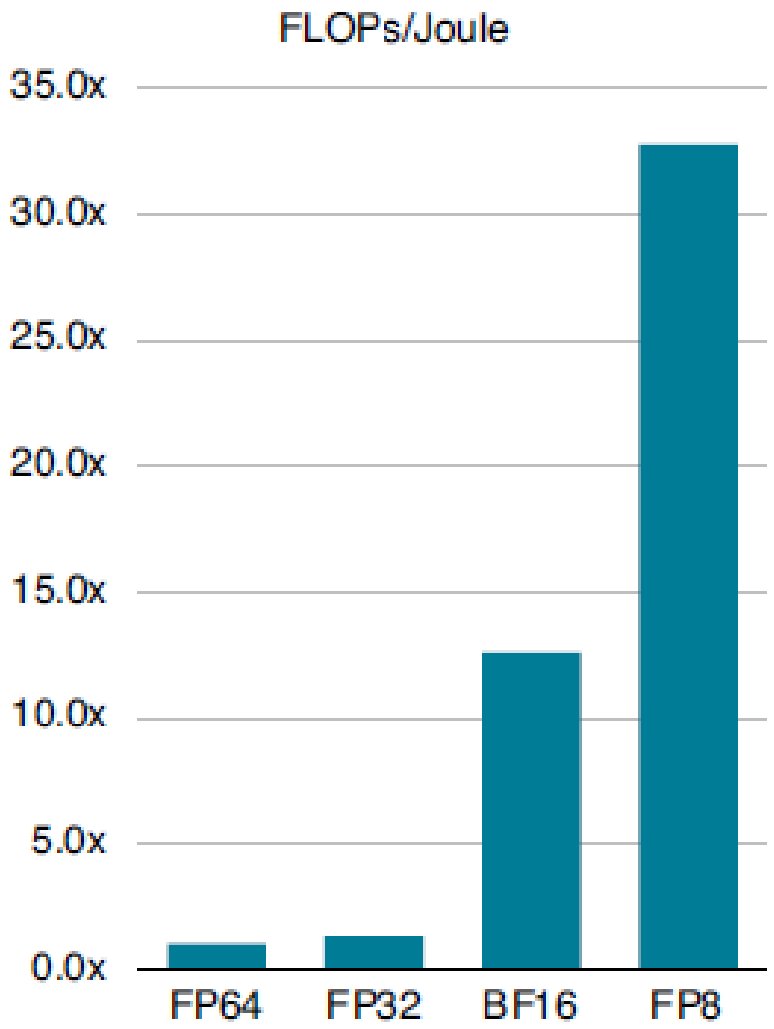
Does not overflow

BFloat 16

1 0 1 0 1 0 0 0 1 0 1 1 0 0 0 0 ~~1 0 0 0 1 1 0 1 0 0 0 0 0 1 1 0~~

Relative costs

Bit-width	Operation	Energy	Relative costs
32-bit	Floating-point **ADD**	0.9pJ	30
	Floating-point **MUL**	3.7pJ	123.33
	Fixed-point **ADD**	0.1pJ	3.33
	Fixed-point **MUL**	3.1pJ	103.33
	DRAM access (Average)	0.65~1.3nJ	21667~43333
16-bit	Floating-point **ADD**	0.4pJ	13.33
	Floating-point **MUL**	1.1pJ	36.67
	*Fixed-point **ADD**	0.05pJ	1.67
	*Fixed-point **MUL**	1.55pJ	51.67
	DRAM access (Average)	0.33~0.65nJ	10000~21667
8-bit	Fixed-point **ADD**	0.03pJ	1
	Fixed-point **MUL**	0.2pJ	6.67
	DRAM access (Average)	0.16~0.33nJ	5333~10000

Zhao et al., Cambricon-Q: A Hybrid Architecture for Efficient Training, ISCA 2021

Selected Floating Point Formats and Associated Energy Efficiency